

# Enes Senel

Senior Scientist, Computational Biology — single-cell multi-omics, machine learning, spatial transcriptomics

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## SUMMARY

Computational biologist with a computer-science background and ten years of experience across machine learning, spatial transcriptomics, and single-cell multi-omics. Currently technical lead for single-cell programs in oncology and immunology at Johnson & Johnson Innovative Medicine, Berlin. Author of open-source methods (Optocoder, novoSpaRc) used by academic and industry labs; co-inventor on a US patent for 3D spatial gene-expression reconstruction.

## EXPERIENCE

### Senior Scientist, Computational Biology — Johnson & Johnson Innovative Medicine, Berlin

*Aug 2025 — Present*

- Lead single-cell multi-omics programs (transcriptomics, proteomics, spatial) that feed biomarker discovery, mechanism-of-action work, and translational strategy for oncology and immunology assets.
- Set the analytical direction across several parallel projects: defining standards, reviewing methodology, and agreeing deliverables with clinical and discovery stakeholders.
- Mentor junior scientists and manage external consultants.
- Work with wet-lab, clinical, and CMC teams to turn analyses into experimental designs and go/no-go decisions.

### Postdoctoral Data Scientist, Computational Biology — Johnson & Johnson Innovative Medicine, Berlin

*Oct 2023 — Jul 2025*

- Built automated Python pipelines for scRNA-seq, CITE-seq, and multi-modal analyses, replacing manual workflows that previously took days per dataset.
- Led the single-cell analyses behind biomarker discovery for a clinical-stage cell therapy and process characterisation for CMC.
- Set up the team's reproducibility conventions — containerisation, version-controlled pipelines, shared QC standards — now the default for new projects.
- Reviewed colleagues' analyses as the team's main technical reviewer.

### Doctoral Researcher, Computational Biology — Max Delbrück Center for Molecular Medicine, Berlin

*Oct 2019 — Sep 2023*

- Designed and led Optocoder, a machine-learning pipeline for decoding barcoded transcripts from imaging-based spatial transcriptomics. Published in NAR Genomics & Bioinformatics.
- Rewrote novoSpaRc (optimal-transport reconstruction of tissue architecture from scRNA-seq) for scale, making it usable on whole-tissue datasets. Co-authored the Nature Protocols paper.
- Applied graph neural networks to spatial omics data, modelling cells as graphs over tissue structure to capture neighbourhood effects that expression-only models miss.
- Co-inventor on a US patent for 3D spatial gene-expression reconstruction.
- Handled the full cycle for both tools: problem framing, method development, benchmarking, release, documentation, and user support.

### Visiting Scientist — The Hebrew University of Jerusalem, Rehovot

*Feb 2023 — Apr 2023*

- Ran an independent short-term project applying protein language models (ProtBERT) to find candidate anti-insecticidal proteins from raw sequence data, combining transfer learning with biological priors.

### Machine Learning Research Engineer — Coriolis Pharma GmbH, Munich

*Apr 2019 — Sep 2019*

- Developed deep-learning models for automated particle detection and classification from flow-microscopy images used in biopharmaceutical QC.
- Packaged ingestion, training, evaluation, and reporting as a Python/PyTorch/TensorFlow tool that internal scientists could run themselves.

### Research Assistant, Computational Neuroscience — Max Planck Institute for Brain Research, Frankfurt

*May 2018 — Sep 2019*

- Applied supervised ML to activity-dependent changes in the neuronal proteome.
- Built a multimodal integration and domain-adaptation pipeline linking proteomic and transcriptomic datasets.

## Undergraduate Research Assistant, Medical Robotics — Ozyegin University Robotics Lab, Istanbul

2014 — 2016

- Developed real-time needle-tip localisation and tracking from ultrasound imaging for autonomous robotic control during biopsy.
- Wrote the C++/CUDA interface connecting real-time image analysis to robot control.

## EDUCATION

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### PhD, Computational Biology (Dr. rer. nat.) — Max Delbrück Center / Humboldt University, Berlin

2019 — 2023

### MSc, Computer Science — Technical University of Munich

2016 — 2019

### BSc, Computer Science — Ozyegin University, Istanbul

2012 — 2016

## SELECTED PUBLICATIONS

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- Benchmarking single-cell foundation models for real-world RNA-seq data integration. bioRxiv (2026).
- Benchmarking Next-Generation Sequencing Platforms: Ultima UG 100 vs. Illumina NovaSeq X Plus. bioRxiv (2026).
- Senel E. et al. Optocoder: decoding barcoded transcripts from imaging-based spatial transcriptomics. NAR Genomics & Bioinformatics (2022).
- Moriel N., Senel E. (co-first) et al. novoSpaRc: flexible spatial reconstruction of single-cell gene expression. Nature Protocols (2021).
- Nitzan M., Karaikos N., Friedman N., Rajewsky N. (incl. Senel E.) Gene expression cartography. Nature (2019).
- Grabarek A., Senel E. et al. Deep learning for particulate impurity detection in cell-based products. Cytotherapy (2021).
- Five robotics papers on ultrasound-guided needle localisation: ICRA 2016, ICRA 2017, ICAR 2015, Mechatronics (2019), TJEECS (2020).
- US patent (co-inventor): 3D spatial gene-expression reconstruction.

## SKILLS

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**Methods:** single-cell multi-omics (scRNA-seq, CITE-seq, spatial), graph neural networks, optimal transport, deep learning, computer vision, protein language models, domain adaptation.

**Engineering:** Python, PyTorch, TensorFlow, scikit-learn, Nextflow, scanpy/scvi-tools, C++/CUDA, containerised reproducible pipelines.

**Leadership:** technical strategy, mentorship, stakeholder management across clinical, discovery, and CMC teams.